

# Dương Hoàng Dũng

Hanoi, Vietnam | +84 968 268 030 | duongdung12a8@gmail.com | <https://github.com/leviance> | <https://www.linkedin.com/in/dũng-dương-95578a208> | <https://delab.id.vn>

## SUMMARY

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I'm a Software Engineer passionate about designing robust architectures, solving complex problems, and building products that empower people and protect their freedom.

Backend and systems engineer with professional experience building performance-sensitive multimedia software, browser-based video infrastructure, cryptographic key-management services, and open-source digital identity platforms. Designed and implemented Milan Player from the ground up using FFmpeg and WebAssembly, achieving approximately 0.5-second live-streaming latency while supporting FLV, HLS, H.264, and H.265 playback across multiple browsers.

Currently leading the research and development of Identra, an open-source Self-Sovereign Identity wallet, while researching distributed trust infrastructure, cryptography, decentralized identifiers, verifiable credentials, and privacy-preserving data exchange.

“Technology is most powerful when it empowers people and protects their freedom.”

## EDUCATION

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### Thai Nguyen University of Information and Communication Technology

Thai Nguyen, Vietnam

*First Prize, University-Level Student Scientific Research Competition, 2020, Winning project: Online learning and video conferencing platform built with Vue.js, Node.js, WebSocket, and WebRTC*

## EXPERIENCE

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### Video Player Engineer — PlayerJS Team, MySoha Platform

November 2021 – February 2024

VCCorp

Hanoi, Vietnam

- Architected and implemented Milan Player, a browser-based multimedia player built from the ground up using FFmpeg, WebAssembly, and HTML Canvas.
- Owned the complete technical implementation, including media ingestion, demultiplexing, decoding, rendering, audio-video synchronization, browser integration, and playback-state management.
- Reduced live-streaming latency to approximately 0.5 seconds, enabling near-real-time playback for latency-sensitive live content.
- Implemented playback support for FLV and HLS/M3U8 streaming formats across multiple browsers.
- Built a customized FFmpeg WebAssembly package of approximately 1.2 MB, supporting H.264 and H.265 video processing while operating within browser resource constraints.
- Designed the player to bypass limitations of native browser media support by performing media processing and decoding through WebAssembly.
- Optimized memory usage, module size, startup behavior, and rendering performance for browser environments.
- Developed cross-browser playback behavior and fallback mechanisms for inconsistent browser capabilities.
- Contributed to the video playback infrastructure used in the VTV.VN ecosystem.
- Investigated and resolved complex issues involving playback latency, corrupted media segments, synchronization drift, browser compatibility, and WebAssembly memory management.
- Collaborated with the PlayerJS team lead on the design of a secure license and key-management system for protected video content.
- Designed and implemented the application layer for a cryptographic key-management architecture using HashiCorp Vault.
- Integrated secure key storage, key rotation, role-based access control, and access audit logging into the video protection workflow.

- Worked closely with the team lead responsible for production deployment, monitoring, and infrastructure operations.

## **Lead Research and Development Engineer**

2024 – Present

*AwnCorp Open-Source Organization*

*Hanoi / Remote*

Leading the technical research, architecture, prototyping, and documentation of open-source systems focused on Self-Sovereign Identity, digital trust, cryptography, and user-controlled digital infrastructure.

## **Lead Architect and Engineer**

*Identra — Open-Source Self-Sovereign Identity Wallet*

<https://identra.id.vn>

- Designed the end-to-end architecture of an open-source SSI wallet for storing, managing, presenting, and exchanging verifiable credentials.
- Developed the product architecture across the web, mobile, and backend layers.
- Built the project website using React.js, Tailwind CSS, and GSAP.
- Developed a mobile application prototype using React Native.
- Designed and implemented backend services using Go.
- Researched architectures for decentralized identifiers, credential issuance, credential presentation, revocation, selective disclosure, secure messaging, and wallet recovery.
- Designed user flows for credential sharing, trusted connections, identity verification, privacy-preserving disclosure, payments, and smart-contract-based exchanges.
- Established technical and product principles that prioritize user ownership, minimal disclosure, interoperability, open-source development, and the avoidance of centralized personal-data collection.
- Produced product requirements, system architecture documents, interface specifications, security models, technical content, and developer-oriented documentation.
- Designed Identra as a non-profit and open-source initiative intended to promote practical SSI adoption within Vietnam.

## **Blockchain Architecture Researcher**

*CertNet — Distributed Trust Infrastructure*

<https://certnet.id.vn>

- Researching the architecture of a hybrid blockchain network designed for high-assurance digital services and Self-Sovereign Identity infrastructure in Vietnam.
- Defined a permissioned-validator, public-read network model in which approved validators participate in consensus while public participants can inspect data, submit transactions, and operate observer nodes.
- Researched Proof-of-Authority consensus, validator governance, public auditability, application-specific chains, rollup architectures, and dual-token economic models.
- Designed an on-chain data strategy centered on decentralized identifier documents, public keys, revocation registries, cryptographic hashes, proofs, and audit records without storing raw personal information.
- Produced technical architecture, governance, token-economics, data-flow, and deployment concepts for the CertNet white paper.
- Evaluated infrastructure requirements for identity systems, digital assets, verifiable records, smart contracts, and national-scale digital trust services.
- Project status: architecture research and concept development.

## **Technical Translator, Editor, and Diagram Designer**

In progress

*Vietnamese Edition of Self-Sovereign Identity*

- Independently managing the complete Vietnamese translation and technical editing of Manning's Self-Sovereign Identity: Decentralized Digital Identity and Verifiable Credentials.
- Translating complex concepts involving decentralized identifiers, verifiable credentials, cryptographic proofs, trust frameworks, selective disclosure, governance, key management, and privacy.
- Established and maintained a consistent Vietnamese technical terminology system for the SSI domain.
- Reworked literal translations into natural, technically accurate Vietnamese suitable for professional publication.
- Redesigned and localized the book's technical diagrams while preserving their logical relationships, visual hierarchy, and original meaning.

- Conducted additional research to update contextual material involving modern identity standards and trust infrastructure.
- Combined software-engineering knowledge, cryptography research, technical writing, and information design throughout the publication process.

## TECHNICAL SKILLS

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**Backend and Systems Engineering:** Node.js, C/C++, concurrent programming, API design, systems architecture, performance optimization

**Web and Application Development:** JavaScript, React.js, React Native, Vue.js, Tailwind CSS, GSAP, WebSocket, WebRTC

**Multimedia and WebAssembly:** FFmpeg, WebAssembly, Emscripten, HTML Canvas, AudioWorklet, FLV, HLS, M3U8, H.264, H.265, audio-video synchronization

**Security and Infrastructure:** HashiCorp Vault, AES-128 content protection, cryptographic key storage, key rotation, access control, audit logging

**Digital Identity and Distributed Systems:** Self-Sovereign Identity, decentralized identifiers, verifiable credentials, selective disclosure, blockchain architecture, cryptography

**Currently Exploring:** goLang, zero-knowledge proofs, distributed consensus, secure messaging, privacy-preserving identity protocols

## CERTIFICATIONS

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English: B1 Certificate

Chinese: HSK Level 2

Vietnamese: Native

## CERTIFICATIONS AND LANGUAGES

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English: B1 Certificate Strong technical reading and listening proficiency; actively developing professional writing and spoken communication.

Chinese: HSK Level 2

Vietnamese: Native

## ENGINEERING PRINCIPLES

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Design systems around clear responsibilities, explicit trust boundaries, and predictable failure modes. Treat performance, security, maintainability, and observability as architectural concerns rather than afterthoughts. Prefer understanding systems from first principles instead of relying exclusively on abstractions. Build technology that gives individuals meaningful control over their identity, information, and digital assets. Approach unfamiliar domains through research, experimentation, implementation, and technical documentation.

## AREAS OF INTEREST

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Backend architecture · Distributed systems · Systems programming · Multimedia infrastructure · WebAssembly · Cryptography · Digital identity · Self-Sovereign Identity · Privacy-enhancing technologies · Blockchain infrastructure · Secure communication · Open-source technology